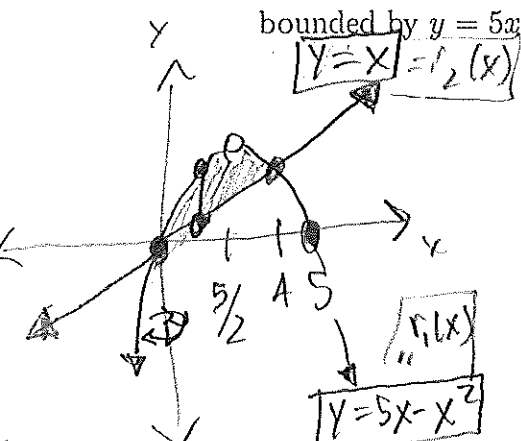


1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



"vertex": $x = \frac{-b}{2a} = \frac{-3}{2(-1)} = \frac{5}{2}$

$$\Rightarrow x=0, x=4$$

$$5x - x^2 = x \Rightarrow x^2 - 4x = x(x-4) = 0$$

③ sample:

$$V_1(x) - V_2(x)$$

$$5y - x^2 - x$$

$$= 4x - x^2$$

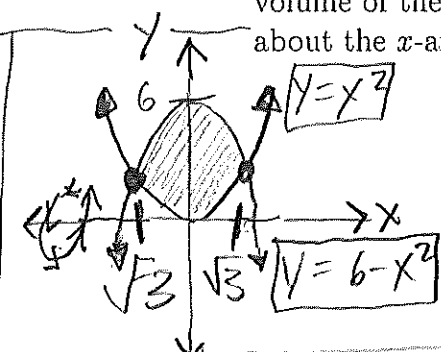
④ compute:

$$V = \int_{-2}^2 (2\pi x)(4x - x^2) dx$$

$$= 2\pi \int_0^4 (4x^2 - x^3) dx = 2\pi \left(\frac{4}{3} x^3 - \frac{x^4}{4} \right) \Big|_0^4$$

$$= 2\pi \left(\frac{4^4}{3} - \frac{4^4}{4} \right) = 2\pi \left(\frac{4^4}{3} - 4^3 \right)$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



① Draw

② Intersections:

② Intersections: $6 - x^2 = x^2 \Rightarrow 2x^2 = 6 \Rightarrow x = \pm\sqrt{3}$

(3) Setu p:

Top: $6 - x^2$, bottom: x^2

$$\Rightarrow V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} ((6-x^2)^2 - (x^2)^2) dx$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$2\pi \int_0^4 x(4x - x^2) dx$$

$$2\pi \int_0^4 4x^2 - x^3 dx$$

$$\left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4$$

$$\frac{4(4)^3}{3} - \frac{(4)^4}{4}$$

$$\frac{256}{3} - 64$$

$$\frac{256 - 192}{3} = \frac{64}{3}$$

$$2\pi \cdot \frac{64}{3} = \frac{128\pi}{3}$$

Handwritten notes for problem 1:

- $5x - x^2 = x$
- $4x - x^2 = x$
- $x^2 - 4x = 0$
- $x = 0, 4$
- $2\pi \int_0^4 x(4x - x^2) dx$
- $80 = 240$
- $8 = 15$
- 255
- $64 \cdot 3 = 180$
- $180 - 12 = 168$
- $168 - 192 = -24$
- $256 - 192 = 64$
- $\frac{64}{3}$
- $2\pi \cdot \frac{64}{3} = \frac{128\pi}{3}$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

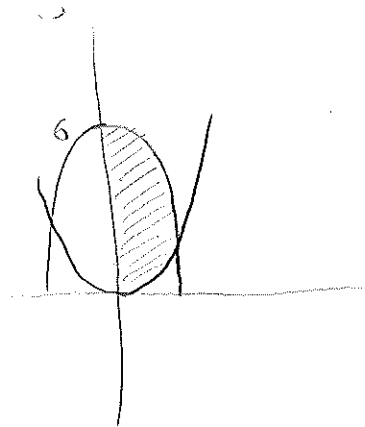
$$\int_0^{\sqrt{3}} \pi (6 - x^2)^2 - (x^2)^2 dx$$

$$6 - x^2 = x^2$$

$$6 = 2x^2$$

$$3 = x^2$$

$$x = \sqrt{3}$$



Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$\int_a^b 2\pi x f(x) dx$$

$$\frac{512}{12} \div 2 = \frac{512}{6}$$

$$y=0$$

$$5(4) - (4)^2$$

$$20 - 16 = 4$$

$$y=4$$

$$\frac{4}{3}(4)^3 - \frac{1}{4}(4)^4$$

$$\frac{4}{3}(64) - \frac{1}{4}(256)$$

$$\frac{256}{3} - \frac{256}{4} = \frac{1024}{12} - \frac{768}{12} = \frac{256}{12}$$

$$5x - x^2 = x$$

$$4x - x^2 = 0$$

$$x(4-x) = 0$$

$$x=0$$

$$4-x=0$$

$$-x = -4$$

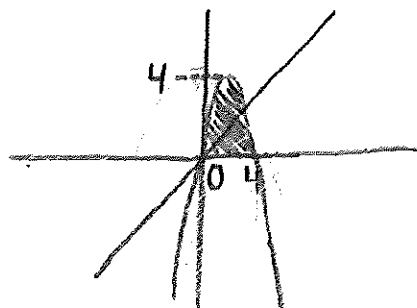
$$x = 4$$

$$2\pi \int_0^4 x(5x - x^2 - [x]) dx$$

$$2\pi \int_0^4 4x^2 - x^3 dx$$

$$2\pi \left[\frac{4}{3}x^3 - \frac{1}{4}x^4 \right]_0^4$$

$$\boxed{\frac{512}{12} \pi}$$



2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$\int_a^b 2\pi (6 - x^2)^2 - (x^2)^2 dx$$

$$2\pi \int_{-\sqrt{3}}^{\sqrt{3}} 36 - 12x^2 + x^4 dx$$

$$36x - \frac{12}{3}x^3 + \frac{1}{5}x^5 \Big|_{-\sqrt{3}}^{\sqrt{3}}$$

$$36\sqrt{3} - \frac{12}{3}\sqrt{3} + \frac{1}{5}(\sqrt{3})^5 - \left(36(-\sqrt{3}) - \frac{12}{3}(-\sqrt{3})^3 + \frac{1}{5}(-\sqrt{3})^5 \right)$$

$$36\sqrt{3} - \frac{39}{3}\sqrt{3} + \frac{9}{5}\sqrt{3}$$

$$36\sqrt{3} - 13\sqrt{3}$$

$$23\sqrt{3}$$

$$\frac{115}{5}\sqrt{3} + \frac{9}{5}\sqrt{3} = \frac{124}{5}\sqrt{3}$$

$$\times 2\pi$$

$$6 - x^2 = x^2$$

$$6 - 2x^2 = 0$$

$$2(3 - x^2) = 0$$

$$3 - x^2 = 0$$

$$-x^2 = -3$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

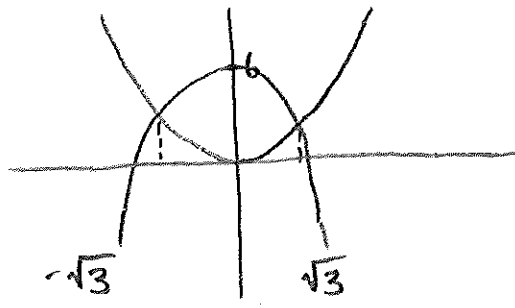
$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

$$x = \pm\sqrt{3}$$

$$x = \pm\sqrt{3}$$

$$x = \pm\sqrt{3}$$



$$\frac{124}{5} \sqrt{3} \pi$$

$$\boxed{\frac{248}{5} \sqrt{3} \pi}$$

Andy Bumpard
1571290

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$\int_0^4 (5x - x^2) - (x) dx$$

$$\int_0^4 (4x - x^2) dx$$

$$\left[2x^2 - \frac{x^3}{3} \right]_0^4$$

$$\left[32 - \frac{64}{3} \right]$$



$$5x - x^2 = x$$

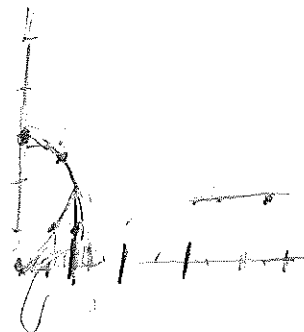
$$4x - x^2 = 0$$

$$x(4 - x) = 0$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$x = \sqrt{6 - y}$$

$$x = \sqrt{y}$$



$$\int_0^6 (6 - x^2) - x^2 dx$$

$$\left[6x - \frac{2}{3}x^3 \right]_0^6$$

$$6 - 2x^2 = 0$$

$$\sqrt{6 - y} = \sqrt{y}$$

$$\sqrt{6 - y} - \sqrt{y} = 0$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

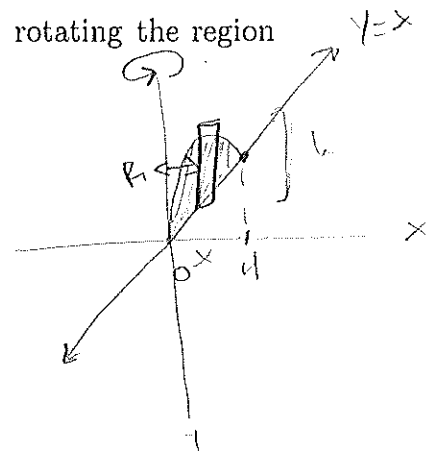
$$-x^2 + 5x$$

x	y
0	0
1	4
2	6
3	6
4	4
5	0

$$x(4x - x^2) \quad \text{terms of } x$$

$$2\pi \int_0^4 x(5x - x^2 - (x)) dx = 4x^2 - x^3$$

$$V = 2\pi \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4$$



$$5x - x^2 = x$$

$$x^2 - 4x = 0$$

$$x(x - 4) = 0 \Rightarrow x = 0, x = 4$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis. Terms of x

$$6 - x^2 = x^2$$

$$6 = 2x^2$$

$$3 = x^2$$

$$x = \pm \sqrt{3}$$

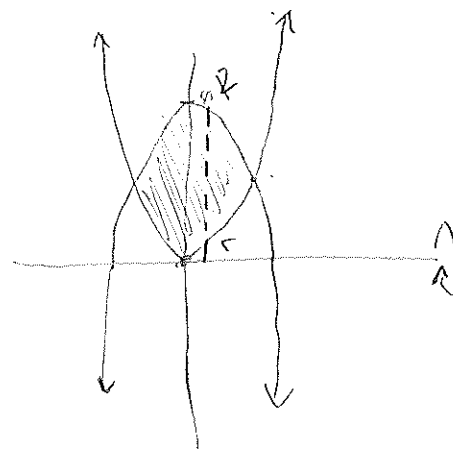
$$\pi \int_{-\sqrt{3}}^{\sqrt{3}} (6 - x^2)^2 - (x^2)^2 dx$$

$$(6 - x^2)(6 - x^2) - x^4$$

$$36 - 6x^2 - 6x^2 + x^4 - x^4$$

$$36 - 12x^2 \quad \frac{12}{2} = 6$$

$$V = \pi \left[36x - 4x^3 \right]_{-\sqrt{3}}^{\sqrt{3}}$$



Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$V = 2\pi \int (outer\ R)^2 - (inner\ R)^2$$

$$2\pi \int_0^4 [(5x - x^2) - x] dx$$

$$2\pi \int_0^4 [-x^2 + 5x - x] dx$$

$$2\pi \int_0^4 [-x^2 + 4x] dx$$

$$2\pi \left[-\frac{x^3}{3} + \frac{4x^2}{2} \right]_0^4$$

$$2\pi \left(-\frac{64}{3} + 32 \right)$$

$$2\pi \left(\frac{32}{3} \right)$$

$$\frac{64}{3} \pi$$

$3^2 \times 3 = 27$
 $16 \times 4 = 64$
 $2^2 \times 4 = 16$
 $9 \times 6 = 54$
 $4 \times 4 = 16$
 $2 \times 4 = 8$
 $1 \times 4 = 4$
 $0 \times 4 = 0$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$x = \sqrt{y}$$

$$x = \sqrt{6-y}$$

$$x^2 = 6 - y$$

$$x =$$

$$2\pi \int [(\sqrt{6-y})^2 - (\sqrt{y})^2] dy$$

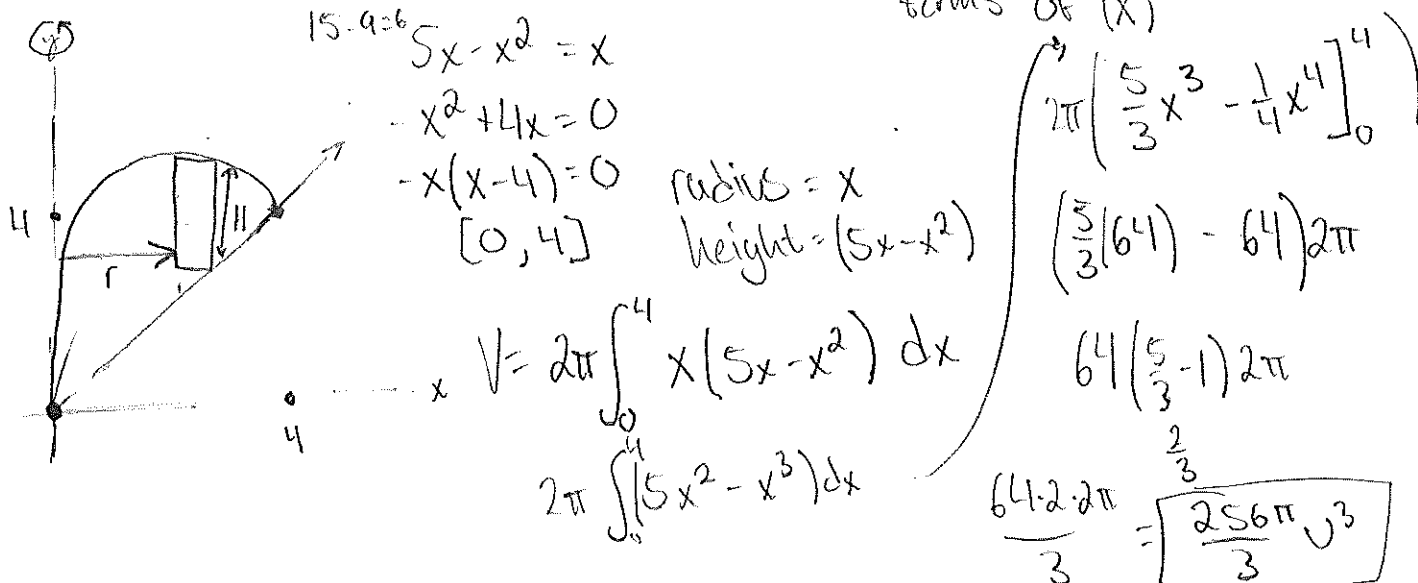
$$2\pi \int [6 - y - y] dy$$

$$2\pi \int [6 - 2y] dy$$

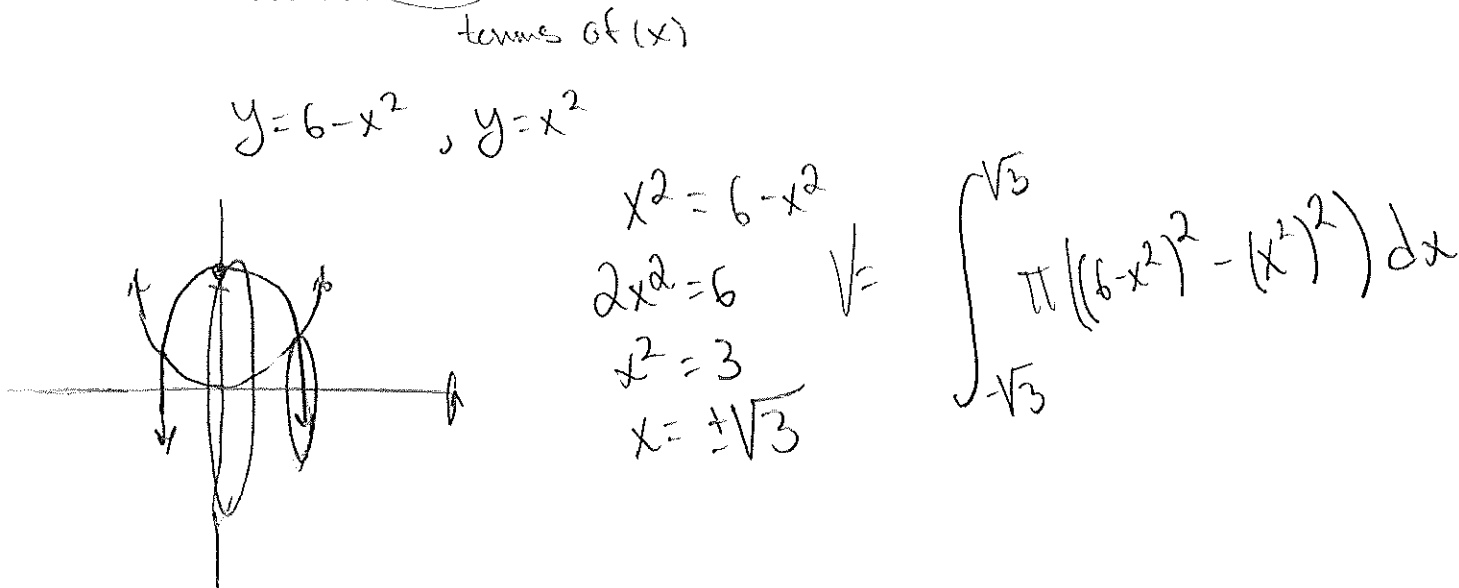
Name: Brett Winter
 ID Number: 1637606

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

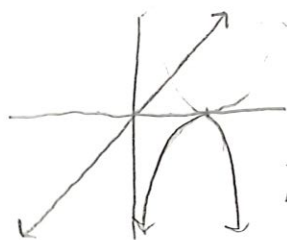


2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



$$x = 5x - x^2$$

$$0 = -x^2 + 4x$$

$$-x(x-4)$$

- respect to x

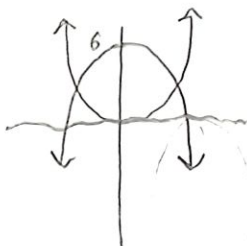
- intercepts at 0, 4

$$2\pi \int_0^4 (x)(5x - x^2 - x) dx$$

$$2\pi \left[\frac{x^2}{2} \left(\frac{4x^2}{2} - \frac{x^3}{3} \right) \right]_0^4$$

$$2\pi \left[\frac{x^2}{2} \left(2x^2 - \frac{x^3}{3} \right) \right]_0^4$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



- respect to x

$$x^2 = 6 - x^2$$

$$0 = 6 - 2x^2$$

$$-2x^2 + 6 = 0$$

$$-2(x^2 + 3)$$

$$\pi \int_{-3}^3 (6 - x^2)^2 - (x^2)^2 dx$$

$$\pi \left[\left(36x - \frac{x^5}{5} \right) - \left(\frac{x^5}{5} \right) \right]_{-3}^3$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$5x - x^2 = x \Rightarrow 4x - x^2 = 0$$

$$\Rightarrow x(4-x) = 0 \quad x = 0, 4$$

$$V = 2\pi \int_0^4 x(4x - x^2) dx$$

$$V = 2\pi \int_0^4 (4x^2 - x^3) dx$$

$$V = 2\pi \int_0^4 \left(\frac{4x^3}{3} - \frac{x^4}{4} \right) dx \Big|_0^4$$

$$V = 2\pi \int_a^b (r)(h)$$

$$\frac{4(4)^3}{3} - \frac{(4)^4}{4} \times 2\pi$$

$$\frac{4(64)}{3} - \frac{(4)^4}{4}$$

$$r = x$$

$$h = 4x - x^2$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$V = \pi \int_a^b (R)^2 - (r)^2$$

$$6 - x^2 = 0 \Rightarrow 6 = x^2$$

$$y = x^2 \Rightarrow x = \sqrt{y}$$

$$6 - x^2 = x^2 \Rightarrow 6 = 2x^2 \Rightarrow x = \sqrt{3}$$

$$V = \pi \int_0^6 ((\sqrt{y})^2)^2 - (6)^2 dy$$

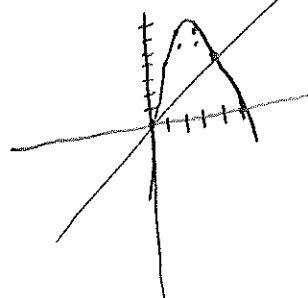
$$V = \pi \int_0^6 (y^2 - 36) dy \Big|_0^6 \Rightarrow \frac{y^3}{3} - 36y$$

flip?

$$\Rightarrow \frac{6^3}{3} - 36(6) \times \pi$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



$$2\pi \int_0^4 x(5x - x^2) dx$$

$x = 5x - x^2$
 $1 = 5 - x$
 $-4 = -x$
 $x = 4, 0$

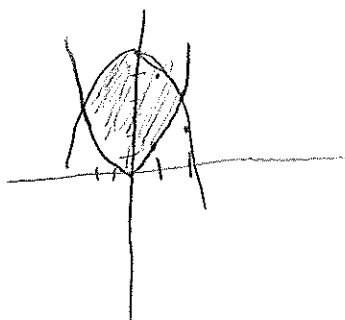
$$2\pi \int_0^4 5x^2 - x^3 dx$$

$$2\pi \left[\frac{5}{3}x^3 - \frac{1}{4}x^4 \right]_0^4$$

$$\frac{320}{3} - 64 = \left(\frac{128}{3} \right) 2\pi$$

$$\frac{256\pi}{3}$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



$$2\pi \int_0^{\sqrt{3}} (6 - x^2)^2 - (x^2)^2 dx$$

$$2\pi \int_0^{\sqrt{3}} (x^4 - 12x^2 + 36) - x^4 dx$$

$$2\pi \int_0^{\sqrt{3}} -12x^2 + 36 dx$$

$$2\pi \left[-4x^3 + 36x \right]_0^{\sqrt{3}}$$

$$6 - x^2 = x^2$$

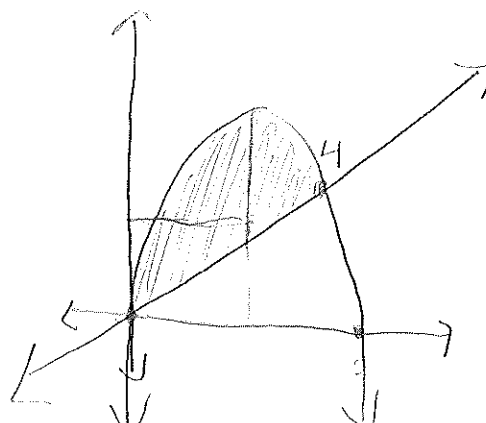
$$6 = 2x^2$$

$$3 = x^2$$

$$x = \pm\sqrt{3}$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



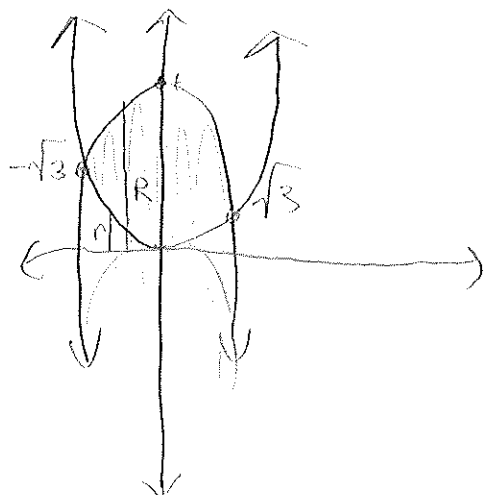
$$\begin{aligned}
 5x - x^2 &= x \\
 4x - x^2 &= 0 \\
 x(4 - x) &= 0 \\
 x &= 0 / 4
 \end{aligned}$$

Shell height: $5x - x^2 - (x) = 4x - x^2$

Shell radius: $x - (0) = x$

$$\begin{aligned}
 V &= \int_0^4 2\pi(x)(4x - x^2) dx \\
 &= \int_0^4 2\pi(4x^2 - x^3) dx \\
 &= 2\pi \left(\frac{4}{3}x^3 - \frac{1}{4}x^4 \right) \Big|_0^4 \\
 &= 2\pi \left(\frac{4}{3}(64) - \frac{1}{4}(256) \right) \\
 &= 2\pi \left(\frac{256}{3} - 64 \right) \\
 &= 2\pi \left(\frac{256 - 192}{3} \right) \\
 &= 2\pi \left(\frac{64}{3} \right) \\
 &= \frac{128\pi}{3}
 \end{aligned}$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



$$\begin{aligned}
 6 - x^2 &= x^2 \\
 2x^2 &= 6 \\
 x^2 &= 3 \\
 x &= \pm\sqrt{3}
 \end{aligned}$$

$$V = \int_{-\sqrt{3}}^{\sqrt{3}} \pi \left((6 - x^2)^2 - (x^2)^2 \right) dx$$

$$4 \int_0^2 (5-x)^2 dx \Rightarrow 25x^2 - 10x^3 + x^4 \Rightarrow \frac{25}{3}x^3 - \frac{10}{4}x^4 + \frac{1}{5}x^5 \Rightarrow \frac{200}{3} - \frac{180}{4} + \frac{32}{5}$$

Math 166

Quiz 2

Name: Ethan Clemenson

ID Number: 1487557

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

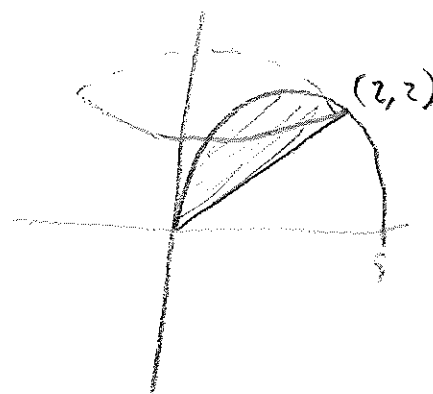
1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$2\pi \int_a^b x f(x) dx$$

$$x = 5x - x^2 \quad 4x = x^2$$

$$2\pi \int_0^2 x(5x - x^2 - x) dx \Rightarrow 2\pi \int_0^2 4x^2 - x^3 dx$$

$$2\pi \left(\frac{4}{3}x^3 - \frac{1}{4}x^4 \right) \Big|_{x=0}^{x=2} \Rightarrow \frac{32}{3} - \frac{12}{3} = \boxed{\frac{40\pi}{3}}$$



2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$\pi \int_a^b f(x)^2 dx \Rightarrow \pi \int_{-\sqrt{3}}^{\sqrt{3}} (6 - x^2 - x^2)^2 dx$$

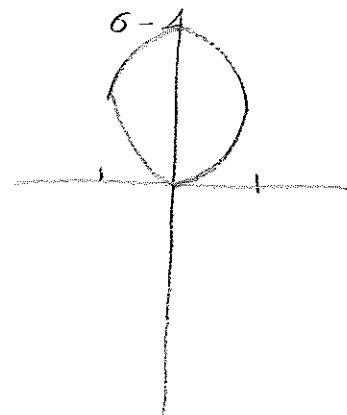
$$6 - x^2 = x^2$$

$$6 = 2x^2$$

$$3 = x^2$$

$$\pm\sqrt{3}$$

$$\int_{-\sqrt{3}}^{\sqrt{3}} (6 - 2x^2)^2 dx$$



$$\int_{-\sqrt{3}}^{\sqrt{3}} 36 - 12x^2 + 4x^4 dx$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$V = \int_{-2}^2 2\pi x (5x - x^2) dx$$

$$5x - x^2 = x$$

$$V = \int_a^b 2\pi x (f(x) - g(x)) dx$$

$$2\pi \int_{-2}^2 (5x^2 - x^3) dx$$

$$5x - x^2 = x$$

$$\hookrightarrow 2\pi \cdot \left[\frac{5x^3}{3} - \frac{x^4}{4} \right]_{-2}^2$$

$$\begin{aligned} x^2 + 1x &= 5x \\ x^2 - 4x &= 0 \\ x - 2, 2 \\ 2, -2 \end{aligned}$$

$$2\pi \left(\left(\frac{40}{3} - 4 \right) + \left(\frac{40}{3} - 4 \right) \right) = V$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$V = \int_0^3 \pi (x^2 - 6x^2)^2 dx$$

$$6 - x^2 = x^2$$

$$x^2 - 6 = -x^2$$

$$2x^2 - 6 = 0$$

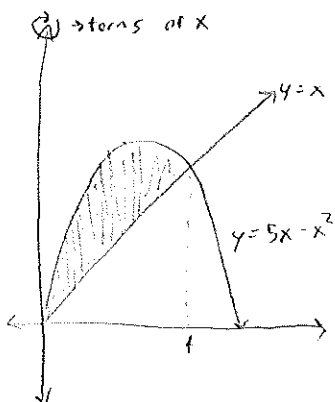
$$2(x^2 - 3)$$

$$x - 3 \quad x = 0$$

$$x = 0, 3$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



$$\begin{aligned} 5x - x^2 &= x \\ -x^2 + 4x &= 0 \\ -x(x-4) &= 0 \\ x &= 4 \end{aligned}$$

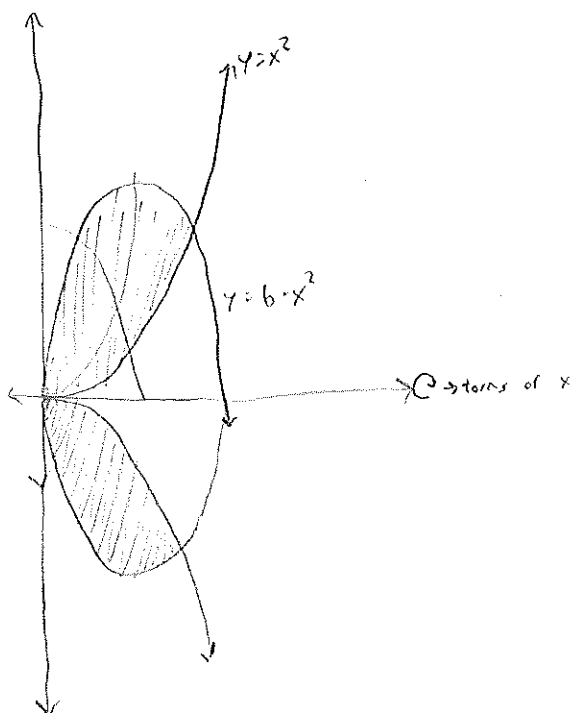
$$V = \int_0^4 2\pi x (5x - x^2 - x) dx$$

$$2\pi \int_0^4 x(-x^2 + x + 5) dx$$

$$2\pi \int_0^4 (-x^3 + x^2 + 5x) dx$$

$$2\pi \left[-\frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{5}{2}x^2 \right]_0^4$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



$$\begin{aligned} 6 - x^2 &= x^2 \\ -2x^2 + 6 &= 0 \\ -2(x^2 - 3) &= 0 \\ x^2 - 3 &= 0 \\ x^2 &= 3 \\ x &= \pm\sqrt{3} \end{aligned}$$

$$V = \int_0^{\sqrt{3}} \pi (6 - x^2)^2 - (x^2)^2 dx$$

$$V = \pi \int_0^{\sqrt{3}} (6 - x^2)^2 - x^4 dx$$

$$V = \pi \int_0^{\sqrt{3}} (2x^4 - x^4 - 12x^2 + 36) dx$$

$$V = \pi \int_0^{\sqrt{3}} x^4 - (2x^2 + 36) dx$$

$$\pi \left[\frac{1}{5}x^5 - 4x^3 + 36x \right]_0^{\sqrt{3}}$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



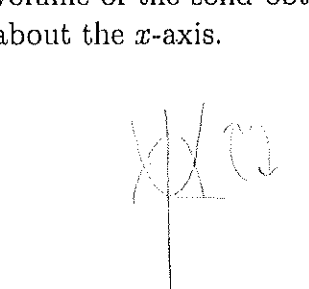
$x_1 = x_2$
 $k: y(x)$
 $y = 5x - x^2$
 $y = x$
 $x: 0$
 $x: 4$

$$2\pi \int_0^4 x (5x - x^2 - x) dx$$

$$2\pi \int_0^4 (4x^2 - x^3) dx$$

$$\left[2\pi \left(\frac{4}{3}x^3 - \frac{1}{4}x^4 \right) \right]_0^4$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



$6 - x^2$
 x^2
 $x^2 = 3$
 $x = \pm\sqrt{3}$

$$\pi \int_{-\sqrt{3}}^{\sqrt{3}} ((6 - x^2)^2 - (x^2)^2) dx$$

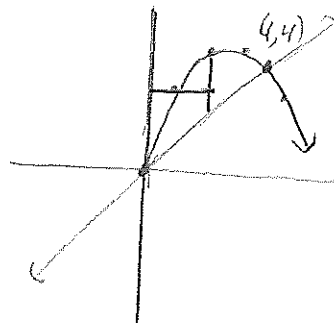
$$\pi \int_{-\sqrt{3}}^{\sqrt{3}} (36 - 12x^2 + x^4 - x^4) dx$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$\begin{aligned}
 & V = 2\pi \int_0^4 (x)(5x - x^2 - x) dx \\
 & = 2\pi \int_0^4 (x)(4x - x^2) dx \\
 & = 2\pi \int_0^4 (4x^2 - x^3) dx \\
 & \boxed{V = 2\pi \left[\frac{4}{3}x^3 - \frac{x^4}{4} \right]_0^4}
 \end{aligned}$$

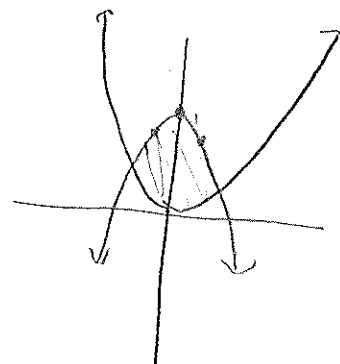
Handwritten notes: $x^2 - 4x + 4$, $x(x-4)$, $x=0$, $x=4$



2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$\begin{aligned}
 & (x^2 + 6)(-x^2 + 6) \\
 & x^4 - 12x^2 + 36 \\
 & V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} [(6 - x^2)^2 - (x^2)^2] dx
 \end{aligned}$$

Handwritten notes: $6 - x^2 = x^2$, $6 = 2x^2$, $\sqrt{3} = x$



$$V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} (x^4 - 12x^2 + 36 - x^4) dx$$

$$V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} (-12x^2 + 36) dx$$

$$\boxed{V = \pi \left[-4x^3 + 36x \right]_{-\sqrt{3}}^{\sqrt{3}}}$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$V = 2\pi \int_0^4 x(4x - x^2) dx$$

$$2\pi \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4$$

$$\frac{256}{3} - 64 = 2\pi \frac{64}{3} = \frac{128\pi}{3}$$

Handwritten work for problem 1:

$$5x - x^2 = x$$

$$4x - x^2 = 0$$

$$x(4 - x) = 0$$

Annotations in the original image:

- 16 (above the 4 in the integral)
- 4 (above the 4 in the integral)
- 16 (above the 4 in the integral)
- 4 (above the 4 in the integral)
- 256 (next to the first term of the antiderivative)
- $\frac{256}{4} = 64$ (next to the second term of the antiderivative)

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

Outer = $6 - x^2$
Inner = x^2

$$V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} \left[(6 - x^2)^2 - (x^2)^2 \right] dx$$

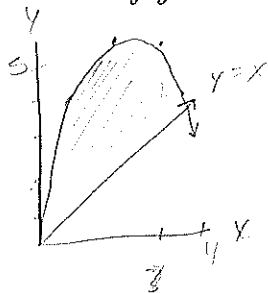
$$6 - x^2 = x^2 \Rightarrow 6 = 2x^2$$

$$3 = x^2 \Rightarrow \sqrt{3} = x$$

Name: Jadon BeeterID Number: 1384273

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



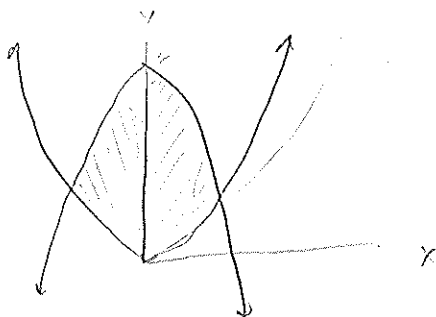
$$5x - x^2 - x$$

$$4x - x^2 = x(4 - x)$$

$$x = 0, 4$$

$$\begin{aligned}
 &= \int_0^4 2\pi x ((5x - x^2) - (x)) dx \\
 &= 2\pi \int_0^4 4x^2 - x^3 dx \\
 &= 2\pi \left(\frac{4x^3}{3} - \frac{x^4}{4} \right) \Big|_0^4 \\
 &= 2\pi \left(\frac{4(64)}{3} - \frac{256}{4} \right) = 0
 \end{aligned}$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



$$\sqrt{6-y} = x \quad x = \sqrt{y}$$

$$(\sqrt{6-y})^2 = (\sqrt{y})^2$$

$$6-y = y$$

$$6-y = y \quad y = 3$$

$$= 2\pi \int_0^3 (\sqrt{6-y})^2 - (\sqrt{y})^2 dy = 2\pi \int_0^3 (6-y) - y dy$$

$$\begin{array}{r} 64 \\ 4 \overline{) 256} \\ \underline{16} \\ 140 \\ \underline{128} \\ 12 \end{array}$$

Math 166
Quiz 2

Name: Jeffrey Dahl
ID Number: 161 32 84

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$y = x(5 - x)$$

$$V = 2\pi \int_0^4 x((5x - x^2) - x) dx$$

$$V = 2\pi \int_0^4 x(4x - x^2) dx$$

$$V = 2\pi \int_0^4 (4x^2 - x^3) dx$$

$$V = 2\pi \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4$$

$$\begin{array}{l} 5x - x^2 = x \\ 4x - x^2 = 0 \\ x(4 - x) = 0 \end{array}$$

$$V = 2\pi \left[\frac{4}{3}x^3 - \frac{1}{4}x^4 \right]_0^4$$

$$V = 2\pi \left[\frac{4}{3}(4)^3 - \frac{1}{4}(4)^4 \right] - 0$$

$$V = 2\pi \left[\frac{4}{3}(64) - \frac{1}{4}(256) \right]$$

$$V = 2\pi \left[\frac{64}{3} \right] \rightarrow V = \frac{128}{3}\pi$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$\begin{array}{l} 6 - x^2 = x^2 \\ 6 = 2x^2 \\ 3 = x^2 \\ \sqrt{\quad} \quad \sqrt{\quad} \\ x = \pm\sqrt{3} \end{array}$$

$$V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} ((6 - x^2) - (x^2)) dx$$

$$V = \pi \int_{-\sqrt{3}}^{\sqrt{3}} (6 - 2x^2) dx$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$\begin{aligned} \text{or } 5x - x^2 &= x & x^2 - 5x + x &= 0 & x &= 5, 0 \\ 5x - x^2 &= x & x(x - 5) &= 0 & & \end{aligned}$$

$$V = \pi \int_0^5 (x)^2 - (5x - 5x^2)$$

$$V = \pi \int_0^5 \frac{x^3}{3} - (-25x^3 + 25x^4) \bigg|_0^5$$

$$V = \pi \int_0^5 \frac{x^3}{3} + \frac{25x^4}{4} + \frac{25x^5}{5} \bigg|_0^5$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$V = 2\pi \int_0^1 x^2 (6 - x^2 - x^2) dx$$

$$V = 2\pi \int_0^1 x^2 (6 - 2x^2) dx$$

$$V = 2\pi \int_0^1 \frac{6x^3}{3} - \frac{2x^5}{5} \bigg|_0^1$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$2\pi \int_0^4 x(5x - x^2 - x) dx$$

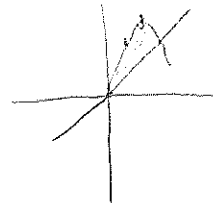
$$4x^2 - x^3$$

$$\left(\frac{4x^3}{3} - \frac{x^4}{4} \right) \Big|_0^4 \cdot 2\pi$$

$$5x - x^2 = x$$

$$x^2 - 4x = 0$$

$$x = 4$$



2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$\int_{-\sqrt{3}}^{\sqrt{3}} 2\pi \int_0^{\sqrt{3}} (6 - x^2)^2 - x^4 dx$$

$$2\pi \int_0^{\sqrt{3}} (6 - x^2)^2 - x^4 dx$$

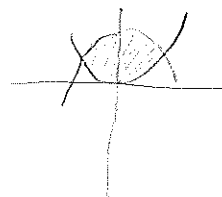
$$36 - 12x^2 + x^4 - x^4$$

$$\left(36x - 4x^3 \right) \Big|_0^{\sqrt{3}} \cdot 2\pi$$

$$6 - x^2 = x^2$$

$$2x^2 - 6 = 0$$

$$x = \sqrt{3}, -\sqrt{3}$$



Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$V = \int_a^b 2\pi (x_R - x_L) dy$$

$$V = \int_0^5 2\pi ((5y - y^2) - y^2) dy = 2\pi \int_0^5 \left(\frac{5y^2}{2} - \frac{y^3}{3} - \frac{y^3}{3} \right) dy$$

$$2\pi \left(\frac{5(5)^2}{2} - \frac{5^3}{3} - \frac{5^3}{3} \right) = \frac{125}{2} - \frac{125}{3} - \frac{125}{3} = 2\pi \left(\frac{125}{2} \right)$$

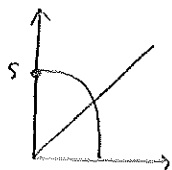
2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$V = \int_0^6 2\pi (x^2 - 6 - x^2) dx$$

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

x -terms



$$5x - x^2 = x$$

$$x^2 - 4x = 0$$

$$x(x - 4) = 0$$

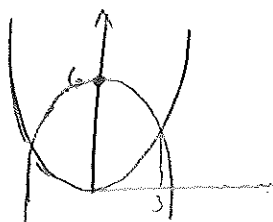
$$x = 0, 4$$

$$2\pi \int_0^4 x(5x - x^2 - x) dx$$

$$2\pi \int_0^4 (5x^2 - x^3 - x^2) dx = 2\pi \left(\frac{5x^3}{3} - \frac{x^4}{4} - \frac{x^3}{3} \right) \Big|_0^4$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

x -terms



$$6 - x^2 = x^2$$

$$6 = 2x^2$$

$$\sqrt{3} = \sqrt{x^2}$$

$$\pm \sqrt{3} = x$$

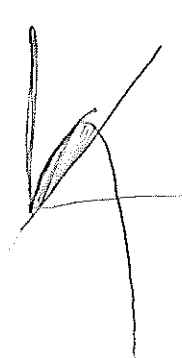
$$\pi \int_{-\sqrt{3}}^{\sqrt{3}} ((6 - x^2)^2 - (x^2)^2) dx$$

$$\pi \int_a^b (f(x)^2 - g(x)^2) dx$$

Name: Max Lockman
ID Number: 163 6753

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.



$$x=0$$

$$x=4$$

$$y = 5x - x^2$$

$$y = x$$

$$4x - x^2$$

$$x(4-x)$$

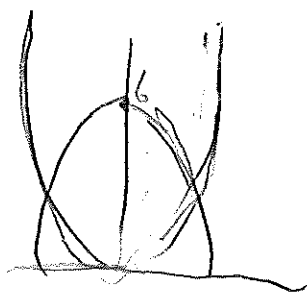
$$2\pi \int_0^4 x(5x - x^2 - x) dx$$

$$2\pi \int_0^4 4x^2 - x^3 dx$$

$$2\pi \left(\frac{4x^3}{3} - \frac{x^4}{4} \right) \Big|_0^4$$

$$2\pi \left(\frac{4^4}{3} - 4^3 \right)$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.



$$y = 6 - x^2$$

$$2x^2 = 6$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

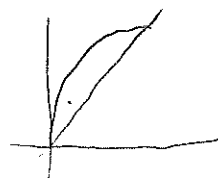
$$\pi \int_{-\sqrt{3}}^{\sqrt{3}} (6 - x^2)^2 - (x^2)^2 dx$$

Name: Joseph HolzerID Number: 1599712

Directions: You will have 20 minutes to complete the quiz individually. You are expected to show all of your work. No calculators, cell phones, or laptops are allowed.

1. Use the shell method to find the volume of the solid formed by rotating the region bounded by $y = 5x - x^2$ and $y = x$ about the y -axis.

$$\begin{aligned} 5x - x^2 &= x \\ y &= x^2 \\ y &= x \end{aligned}$$



$$2\pi \int_0^4 x(5x - x^2) dx$$

$$4x^2 - x^3$$

$$\frac{4}{3}x^3 - \frac{x^4}{4}$$

$$\frac{256}{3} - 64$$

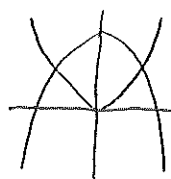
$$\frac{256}{3} - \frac{192}{3} =$$

$$\frac{64}{3} \cdot 2\pi =$$

$$\boxed{\frac{128\pi}{3}}$$

2. Using the washer method, set up but do not evaluate an integral representing the volume of the solid obtained by revolving the region bounded by $y = 6 - x^2$, $y = x^2$ about the x -axis.

$$\pi \int_{-\sqrt{3}}^{\sqrt{3}} (6 - 2x^2)^2 dx$$



$$6 - x^2 = x^2$$

$$\pm\sqrt{3} = x$$